Counting Reports or Counting Evidence:

What Provenance an LLM Orchestrator Needs
to Discount Correlated Agent Reports

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Project direction: Provenance-aware aggregation of agent reports.
Repository: Code, data and project materials.

Project summary

Multi-agent systems often give one model, the orchestrator, the reports of several worker agents. Workers that share a retriever or a tool can pass on the same source. In 40 cases with invented facts, a separate reporter model rewords each source and the accepted reports are frozen. Three judges gave an answer and a probability for every option in 20,280 trials. With nothing disclosed, two reports of a weak source beat one report of a strong source in 454 to 479 of 480 trials per judge. Told that two reports share a source, every judge raised its stated probability of the truth in at least 37 of 40 cases, yet stayed wrong in most trials. Told which source is the primary record, every judge was right in every trial. Even "somewhat more reliable" was enough in 331 of 360 subset trials. Every note we showed was true, so this does not establish robustness to false quality labels.

1 Introduction

Many language-model systems split a task across worker agents and give one model, the orchestrator, the job of combining what the workers report. Workers in one framework often share a retriever or a tool, so their reports can be correlated by construction. Consider three workers that receive the same document and each summarise it in their own words. The reports look like three witnesses, but only one source exists.

A multi-agent system holds the information that could prevent this failure, because it knows which agent called which tool and which document came back. That record is lost when the workers' outputs are rendered as text for the orchestrator. Our research question is the following.

> *Can the orchestrator distinguish the number of agreeing reports from the number and quality of the independent sources behind them, and what must the inter-agent message tell it (a source ID, a source type, that reports share a source, or which source is more reliable) before its belief and its decision follow the evidence?*

We make four contributions. We compare disclosures on identical frozen reports, measure belief and choice side by side with an abstention arm, build a corpus of independently generated paraphrases with provenance kept out of the text, and adapt an index that estimates how many independent sources a judge behaves as if it saw. "Belief" here means stated probability, not calibrated or internal belief. Success means interpretable evidence about the primary hypotheses, including negative results.

After the TA's feedback, we reworded the question to name the aggregation step and the lost source record. The design was unchanged. Post-pilot changes to the screen and abstention controls are documented in Appendix D.

2 Related work

Xu et al. [14] separate conflicts between supplied evidence from conflicts with a model's own knowledge. Our setting is the first kind, while ClashEval studies the second [13]. We remove that axis by using invented entities throughout. Schuster et al. [10] find a credibility hierarchy that repetition can reverse. Their repeated evidence is verbatim. RAMDocs gives each document its own agent, so count and independence cannot be separated [11]. AuthorityBench studies authority from URLs [16].

Multi-agent debate treats agreeing outputs as corroboration [3]. Minority Sentinel trains a classifier to overturn wrong majorities [6], while conformity and sycophancy can push debate toward consensus [1, 15]. Our agents never interact, and their agreement is inherited from the source. Provenance-aware memory methods correct the count outside the model [8, 9], and provenance through paraphrasing remains an open challenge [12]. We measure what the judge itself does when nobody tells it, and once it is told in one sentence. Length and influence studies motivate our controls [5, 7]. Our independence index adapts correlation neglect, the failure to account for repetition in one's information [2, 4].

3 Study design

Cases and frozen reports. Each case fixes an invented world truth, three strong sources B, C and D that state it through separate routes, one weak source A that states a wrong value, and two distractors that no source supports. The judge chooses among these four options, separating following A's claim from arbitrary mistakes. The 40 cases cover institutional logistics (13), product specifications (13) and incident status (14), with ten per quality dimension: authority, directness, method and primary-source status. In 30 cases (T1), A's sentence carries a mild hedge that survives rewording. In ten (T2), A and B are equally flat assertions. The two tiers show whether a result needs such a content cue.

A reporter model, Gemma 4 31B, rewrites each source's content sentence and never sees who wrote it. Candidates must keep meaning and values, pass length and overlap checks, and contain no attribution. Claude Sonnet 5 checks semantic equivalence. We freeze 280 reports and reuse them across conditions. Members reviewed the cases. One member read every frozen report before

judging. In seven cases, one of A's reports echoed the question, so we report results with and without these cases. Appendix A gives the construction and acceptance rules.

Two complementary blocks. In the agreement block, every report supports the truth. A0 has no report and A1 has one. A2 joins three rewordings of B in one message, A3 separates them into three lines from one agent, and A4 assigns three agent names. A5 adds a shared source ID, and A6 states their dependence. A7 uses three independent sources without disclosing their independence. A message is one labelled report line in a prompt, not a live tool call.

In the conflict block, two rewordings of weak A face one report of strong B, so counting reports gives the wrong answer. Table 1 lists the disclosures. Report content and count stay fixed. Full conditions and example prompts appear in Appendices A and E.

Table 1: Conflict conditions. The same two weak-source reports and one strong-source report appear in every condition. All disclosed information is correct.

Condition	Information added to the reports
X1	None.
X2	Each report's source type, such as a student guide or an official register.
X3	The two agreeing reports share a source. B's report has a different, independent source.
X4	B's source is a primary, authoritative record, while the others use secondhand or unofficial material.
X5	Both the dependence and quality notes.
W1–W3	X3 plus "somewhat", "considerably" or "far more reliable" for B's source.

Models, repetitions and study stages. The judges are Claude Opus 5, Claude Sonnet 5 and GPT-5.4, called through OpenRouter. Each is pinned to one serving provider. They return an answer and integer probabilities per option, summing to 100. Reasoning was switched off explicitly, and every main-run trial logged zero reasoning tokens. The runner sends temperature 0, but no endpoint accepts it, so each judge ran at its provider's default.

Stage 1 ran 17,760 trials on 40 cases, with six repeats per agreement condition and 12 per conflict condition for each judge. The conflict schedule balances the truth's option position and the independent report's slot. The confidence screen and abstention checks use four rotations per case. Stage 2, two days later, ran 2,520 trials on ten preselected cases: the wording ladder for all three judges and 88–96-word reports for Sonnet and GPT-5.4. Every domain and quality dimension appears, and three cases are T2. The 3,420-trial pilot and optional checks are separate from the 20,280 main-study trials. Appendix A gives trial accounting.

4 Measures and analysis

We ranked the measures in advance, because they can disagree. The decision against the world truth is primary, followed by decision stability, stated $P(\text{truth})$ and abstention. Stated probabilities are behavioural outputs, not calibrated confidence or direct observations of internal belief. We

compare them within judge. The confidence screen requires rising $P(\text{truth})$ across an irrelevant report (P0), a sentence ruling out two options (P1), and a report naming the answer (P2), using Page's ordered-alternatives test with cases as blocks.

With $l(c)$ the logit of a case's mean stated $P(\text{truth})$ in condition c,

$$n_{\text{eff}}(c) = \frac{l(c) - l(\text{A0})}{l(\text{A1}) - l(\text{A0})}. \tag{1}$$

The index expresses the update relative to one report. Values of 1 for repeated evidence and 3 for three independent sources are reference values under equal diagnostic strength and additive log-odds, not unconditional norms. We compute it per case and report medians. It is undefined for one-report updates below 0.2 logits, and means outside 1–99% give bounds. Saturation limits interpretation, especially for GPT-5.4.

The six hypotheses predict repetition effects (H1), effects of redundancy and segmentation rather than agent names (H2), confidence changes without decision changes under dependence disclosure (H3), decision changes under quality disclosure (H4), benefits from named source types (H5), and abstention where the judge's probabilities weakly support its answer (H6). H1 and H2 follow prior work. H3, H4 and H6 are primary targets. The hypotheses and interpretation criteria were agreed before data collection. Appendix D lists procedural changes after the pilot.

A case's repeats use the same reports and are not independent evidence. We therefore use cases for sign tests and 95% bootstrap intervals from 10,000 case resamples, reported per judge. These describe variation within this authored corpus, not a random sample of real agent tasks. Nonsignificant tests do not establish equivalence. Appendix A.3 gives the full measurement rules.

5 Results

All three judges passed the confidence screen (Page's $p < 0.0001$). One stage-1 reply failed to parse and 18 more were imperfect. All 19 fell in no-evidence or screen conditions, leaving A1–A7 and conflict counts unaffected. All stage-2 trials parsed cleanly. Conflict decisions were stable: each cell's most common answer appeared in 96–97% of its 12 repeats on average, against 41% by chance. The screen supports probability order and differences, but does not establish calibration. A discrimination gate removed after the pilot would have excluded Sonnet and GPT-5.4 from probability-based results.

5.1 Repetition increased confidence without new sources

Three reports of one source raised every judge's belief (Table 2). From A1 to A4, mean $P(\text{truth})$ rose in 40 of 40 cases for every judge and in all three domains (H1). For Opus and Sonnet, most of the rise came from splitting the rewordings into separate report lines (H2). Naming three agents added less. GPT-5.4 was near its ceiling from A2 on.

Table 2: Agreement block: mean stated probability of the truth (%), across 40 cases and six rotations per case. A1–A6 reuse one source. A7 contains three independent sources whose independence is not disclosed.

Condition	Opus	Sonnet	GPT-5.4
A1: one report	77.4	76.4	91.1
A2: three rewordings, one message	81.2	84.2	96.8
A3: three report lines, one agent	91.4	94.8	99.6
A4: three report lines, three agents	93.8	95.2	99.5
A5: A4 plus shared source ID	88.3	95.0	99.8
A6: A5 plus dependence note	85.6	91.3	99.6
A7: three independent sources	94.4	95.8	99.5

Three independent sources (A7) earned the same belief as A4 to within a point. With provenance hidden, the judges had no way to tell these rewordings from independent reports. For A4 versus A7, median n_{eff} was 1.64 versus 1.69 for Opus and 1.88 versus 1.98 for Sonnet. GPT-5.4's index was at ceiling in 35 of 40 cases. Opus's compressed upper scale also limits comparison with the reference value of 3. The middle of the scale gives a cleaner reading. With A6's dependence note, three reports of one source still counted as 1.24 sources for Opus and 1.52 for Sonnet. The correction was incomplete under the index's assumptions, but probability contrasts are stronger evidence than literal source counts.

5.2 Dependence disclosure shifted confidence but rarely resolved conflict

With nothing disclosed, the judges were right in 1, 2 and 26 of 480 trials, and every wrong answer named A's claim (Table 3). The note in X3 raised mean P(truth) by 15.5 points for Opus (95% CI 14.8–16.3), 12.3 for Sonnet (11.7–13.0) and 7.3 for GPT-5.4 (5.7–8.7), in 40, 40 and 37 of 40 cases. Yet every judge still put more probability on A's claim than on the truth on average.

Table 3: Conflict block: correct decisions and mean stated P(truth) (%). Each cell covers 40 cases and 480 trials per judge. A correct quality note resolves every tested conflict. Dependence alone leaves most decisions wrong.

	Correct / 480			Mean P(truth) (%)		
	Opus	Sonnet	GPT-5.4	Opus	Sonnet	GPT-5.4
X1: no disclosure	1	2	26	25.7	20.2	22.6
X2: source types	403	193	283	60.9	41.1	52.0
X3: dependence	63	8	27	41.2	32.5	29.9
X4: quality	480	480	480	68.9	74.3	82.0
X5: both	480	480	480	68.8	67.3	75.8

H3's belief half held for every judge, but its prediction of unchanged decisions did not hold uniformly. Opus improved in 14 cases and worsened in none ($p < 0.0001$). All 63 right X3 answers fell in T1 cases, where A's hedge already leaned toward the truth. Sonnet improved from 2 to 8

right answers, and GPT-5.4 from 26 to 27. Their case-level tests ($p = 0.0625$ and $p = 0.377$) provide insufficient evidence of a systematic increase, not proof of no effect. In the ten T2 cases, every judge scored 0 of 120 with or without the dependence note. The note raised belief but rarely changed decisions.

5.3 Correct quality information consistently changed decisions

With the quality note (X4) or both notes (X5), every judge was right in all 480 trials (H4). Source-type names (X2) also helped, by an amount that depended on the judge (H5): 403, 193 and 283 right decisions. The gain remained when cases naming familiar public or official bodies were excluded. It also varied by quality dimension (Appendix C).

The wording ladder separates decisions from belief (Figure 1). At "somewhat more reliable", Opus and GPT-5.4 were right in all 120 trials and Sonnet in 91. Mean P(truth) sat near 50–54%, so the judges chose the better source while the repeated claim kept much of their belief. From "considerably" on, every judge was right in every trial, and belief rose further with "far". All 29 Sonnet errors named A's claim, 16 in the three T2 cases. X5 did not produce the highest confidence. Decisions changed under mild wording, ruling out a requirement for the strongest wording. Every note favoured the truth, however, so this result cannot rule out simple compliance.

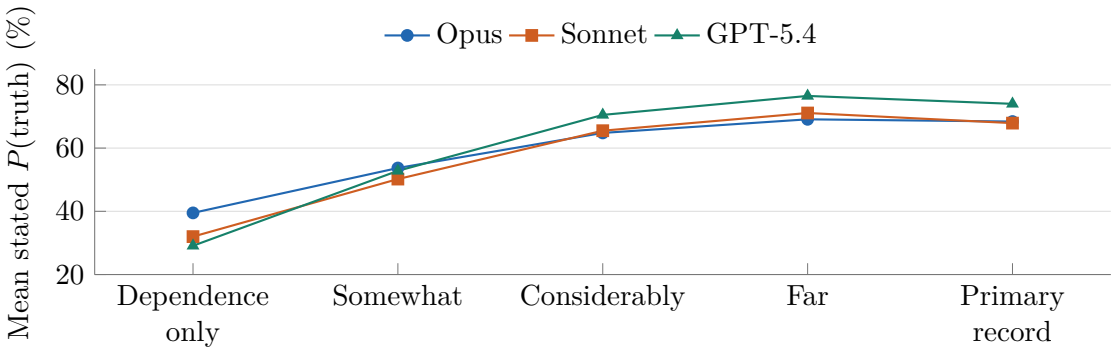


Figure 1: Wording ladder on ten cases, with 120 trials per judge and condition. Every rung keeps the dependence note. X3 and X5 are stage-1 results for these cases. The three comparative wordings ran two days later in stage 2. Lines connect discrete conditions. Case-resampling intervals are at most ±2.0 percentage points.

5.4 An abstention option rarely exposed uncertainty

Few forced-choice answers were unbacked by the judge's own numbers. H6 flags an answer below the highest-probability option or a gap below ten points between the top two probabilities, after rescaling to 100. Only 30, 3 and 7 of each judge's 2,400 conflict records were flagged, all close calls. Only six had twins in the four-repeat abstention arm. None of the six abstained. This direct test is too sparse to rule H6 out generally.

All three judges can abstain. Shown one irrelevant report, they abstained in 160, 153 and 160 of 160 trials, and none abstained in A7. Yet Opus and Sonnet never abstained in their 800 conflict trials, and GPT-5.4 abstained in 9 of 800. H6 therefore found no support. The errors usually came

with substantial stated support for the repeated claim, so an abstention option did not remedy them.

6 Robustness and scope of the evidence

Longer reports. With 88–96-word reports (Sonnet and GPT-5.4, ten cases), the conflict block broadly matched the short-report results. The dependence note raised $P(\text{truth})$ in all ten cases for both judges. GPT-5.4 was right in 18 of 120 X3 trials, against 3 with short reports, all in T1 cases. With either quality note, both judges were right in every trial. Sonnet's A1-to-A2 rise was 13.4 points with long reports against 8.0 with short ones, larger in nine of ten cases ($p = 0.011$). This comparison crosses days, so it cannot isolate a causal length effect. GPT-5.4 had little headroom: one long report already gave it 94.8% $P(\text{truth})$.

Targeted supplementary checks. With the seven echo cases left out, every direction held. On new reports for two cases, X1/X3 failures and X4/X5 successes held, while X2 improved. Sonnet made identical decisions with reasoning on and off in a two-case comparison. Only 32 of 144 enabled trials produced reasoning. All 20 X3 traces acknowledged the shared source, yet chose the repeated claim. This shows failure after acknowledgment on those trials, not a general test of reasoning models. With long reports on nine cases, Sonnet's median n_{eff} was 1.95 for rewordings and 1.97 for independent sources.

The token-probability cross-check used two models that return token probabilities, because the main judges do not. Both readings ordered the screen's three levels alike and correlated across trials (Spearman 0.83 for Llama 4 Maverick and 0.81 for Qwen 3.6 27B). They part with irrelevant evidence: Qwen stated 25% on every option while its tokens put 77% on one number on average. This supports ordinal sensitivity on those models, not calibration or validation of the main judges' internal confidence. Details are in Appendix C.

7 Discussion and limitations

What the message should carry. Of the three fields tested in the conflict block, only a quality note made every judge right. Source-type names gave a gain that varied with the judge and quality dimension, and a dependence note moved belief and few decisions. A fourth field, an opaque source ID, ran only in the agreement block. The fields also differ in what they cost. A dependence note needs only what the system already records about which reports share a source. A quality note needs a judgment of which source is better, which is harder to produce and can be wrong.

Taken at face value, X3's note turns two-against-one into one source against one, with nothing said about which is better. The note raised the probability of the truth, but the repeated claim kept more support on average. Every quality note we showed was true, and we never tested a false one. If a false note is followed as readily, whoever writes the field could steer the orchestrator's choice. The practical recommendation therefore depends on independently established quality information.

Limitations. The judge reads fixed text in a single call. Real pipelines pass messages over several steps and tool calls, so we cannot say whether a judge-side fix changes end-to-end accuracy. Each agent reads exactly one source, reports overlap fully rather than partially, and every case is an invented single fact with four options. Every report is a rewording by one reporter. The equivalence rater, Claude Sonnet 5, is also a judge, and two judges come from one developer.

The judges ran at their providers' default temperature, and comparisons between stages carry day-to-day drift. The ladder and length arm used ten cases, the optional checks two to fifteen, and H6's direct test had six flagged twins. The main judges expose only stated numbers, whose upper scale limits the independence index. The prompt asks for the answer before the probabilities, which may tie the numbers to the answer. No control added neutral text of the same length, so generic caution from added words cannot be excluded.

Costs and reproducibility. Logged API spend came to US$52.19 (Appendix D.1). Code, cases, frozen reports, trial logs and setup instructions are in the repository. The archived model identifiers and provider settings specify the versions used.

The study has no human subjects and no personal data. Every entity is invented, so no output can misinform about a real institution, product or incident. Claude Code was used extensively to record decisions, draft documentation and maintain the project's knowledge graph, and as an implementation assistant. The report was drafted with Claude's help, then proofread and edited by all team members.

8 Conclusion

We asked what an orchestrator must be told before it stops counting reports and starts counting evidence. For the three judges we tested, learning that two reports share a source was not enough, and learning which source is better was. Where relative quality is independently known, the message should carry it. Future work should test false quality notes, mild notes without dependence, real tool-call messages and several aggregation steps, with more reporters and judges.

Contribution statement

The contributions below reflect project records and subsequent report revisions. Each member should confirm and update their entry, including any additional report and presentation work, and enter an approximate share before submission. The five shares must sum to 100%.

Member	Recorded contributions	Share
Ong Jia Xi (A0276092Y)	Led the project and ran paid experiments. Completed final case reviews and merges, reviewed all frozen reports, and authored M1 and M38–M40. Wrote the first draft of the final report.	____%
Lee Kai Rong (A0287723U)	Authored nine cases (M5 and M30–M37) and reviewed 17 cases (M4, M8–M9, M16–M19, M22–M28 and M38–M40). Coordinated the corpus and ten-case subset, led the discrimination pass-mark decision, and raised post-pilot changes to the screen, ceiling rule and H6 interpretation. Supported drafting the final report.	____%
Kathleen Pang Qi Yu (A0282219A)	Authored M4 and M22–M29 and reviewed M3, M14–M21 and M38–M40. Wrote the hedge-counting tool and drafted the revised research question.	____%
Ee Rui Kai Rachel (A0281201X)	Authored M3 and M14–M21 and reviewed M2 and M6–M13. Led the wording-ladder sentence review.	____%
Crystal Lai Qiaohui (A0282431L)	Authored M2 and M6–M13 and reviewed M5 and M30–M37. Led the H6 margin-cutoff decision.	____%
Total		100%

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A Detailed protocol and experimental design

A.1 Corpus construction and report acceptance

The 40 cases cover institutional logistics (13), product specifications (13), and incident or status reports (14), with ten cases for each of four quality dimensions: authority, directness, method, and primary-source status. Each case fixes a world truth, a wrong claim from weak source A, three strong sources B, C, and D with distinct evidential routes, and two unsupported distractors. Thirty cases (T1) give A a mild hedge, while ten (T2) give A and B equally flat assertions. The latter remove this particular textual cue to source quality.

Five members authored the cases under 20 rules in `docs/authoring-rules.md`. These require comparable specificity and scope, neutral question wording, no time markers, provenance separated from the content sentence, and distinct evidential routes for B, C, and D. From M6 onward, A's and B's type names differ only along the case's assigned quality dimension. The five pilot cases retain their original names. A static linter checked machine-testable rules, and every case from M2 onward had a reviewer other than its author.

Gemma 4 31B (`google/gemma-4-31b-it`) generated reports without seeing provenance. Each round sampled six candidates at temperature 0.9, increasing to 0.95 in a second round. A candidate had to satisfy all four acceptance requirements:

1. Claude Sonnet 5 marked all details retained, nothing added, all values unchanged, and scope or denominator unchanged, with an equivalence score of at least 80/100.
2. Word-bigram Jaccard overlap was at most 0.70 with the source, and at most 0.50 with another retained report of that source or a source appearing in the same trial.
3. The report contained neither a word unique to its source's type name nor an attribution phrase such as "according to".
4. Its length differed from the source sentence by no more than three words.

Failed candidates were regenerated, not manually edited. Two unsuccessful rounds stopped generation. The provenance-leak check ran again over the frozen reports before judging.

Generation used 1,956 calls for 160 sources. Of 972 candidates, 721 passed the individual checks. We retained 280 reports, comprising two for A, three for B, and one each for C and D per case. The other 441 were surplus or failed pairwise selection. Only two sources needed a second round. Rejection counts, which can overlap, were 105 for length, 72 for source overlap, 66 for overlap with another source's report, and 12 for equivalence. A member read every frozen report before judging. All T1 hedges survived and no case was excluded. Seven cases had a report that echoed the question's wording: M3, M9, M13, M23, M28, M38, and M40. Their exclusion is a sensitivity check, not a change to the corpus.

A.2 Condition definitions and counterbalancing

The agreement block repeats strong source B, so changes in stated confidence can be read on the truth. The conflict block places two reports of A against one of B. Source letters A–D identify authored sources, while the displayed agent letters identify rotating report slots. "Message" denotes

a separately labelled line inside one user prompt, not a separate API message or a live worker interaction.

Table 4: The 13 conditions and the wording ladder. Agent labels (Agent A, B, C) name report slots and rotate with the order, and they are unrelated to the source names A–D. A message is one "Agent ... says" line of the single prompt. Appendix E shows example prompts in full.

	Reports	Messages	Agents	Sources	What the judge is told besides the reports
A0	0	0	0	none	nothing. The question alone gives the judge's prior
A1	1	1	1	B	nothing
A2	3	1	1	B	nothing. Three rewordings of B are joined into one message
A3	3	3	1	B	nothing. Lines labelled "Agent A, report 1" to "report 3"
A4	3	3	3	B	nothing
A5	3	3	3	B	an opaque source ID, "(source: S-001)", on every report
A6	3	3	3	B	A5, plus a dependence note: all "obtained their information from the same underlying source (S-001)"
A7	3	3	3	B, C, D	nothing
X1	3	3	3	A, A, B	nothing
X2	3	3	3	A, A, B	each report's source type name
X3	3	3	3	A, A, B	a dependence note: the two A reports "obtained their information from the same underlying source", while B's "comes from a different, independent source"
X4	3	3	3	A, A, B	a quality note: B's source "is a primary, authoritative record; the other reports rest on secondhand or unofficial material"
X5	3	3	3	A, A, B	X3's note, plus a quality note: B's source "is a primary, authoritative record; the source shared by" the other two "is a secondhand or unofficial one"
W1–W3	3	3	3	A, A, B	X3's note, plus a softened quality note: B's source "is somewhat / considerably / far more reliable than the source shared by" the other two

The conflict block's 12 repeats pair every answer position with every independent-report slot once, and every answer position with every wrong-claim offset once. Agreement uses six repeats under the same fixed schedule in every condition. Abstention uses four repeats, placing the truth in each answer position once. These repeats measure response stability and reduce position imbalance. They are not independent cases.

The length arm embeds each report in one of three authored sets of neutral filler. Filler is assigned by slot, not source, and does not mention an answer option. Each long report has 88–96 words. The longest within a trial is at most 1.10 times the shortest. The report sentence rotates among the start, middle, and end. The ten-case subset is M1–M6, M14, M22, M30, and M38. The selection rule predates the pilot: every domain appears at least three times, every quality dimension at least twice, and three cases are T2.

The abstention arm adds "Cannot be determined from the evidence provided" to X1–X5, A7, and P0. P0 is an irrelevant report used to check whether the judge can abstain. A7 checks whether it

abstains despite three agreeing independent sources. The pre-main-run cut-offs flag fewer than half of P0 trials abstaining as "abstains nowhere" and at least half of A7 trials abstaining as "abstains everywhere".

Table 5: What ran. The short arm and the length arm each have an agreement entry and a conflict entry. We fixed the design of each optional sub-experiment before it ran.

Entry	Conditions	Repeats	Judges	Cases	Trials
Main run, stage 1					
Agreement	A0–A7	6	all three	40	5,760
Conflict	X1–X5	12	all three	40	7,200
Abstention	X1–X5, A7, P0	4	all three	40	3,360
Confidence screen	P0, P1, P2	4	all three	40	1,440
Main run, stage 2					
Wording ladder	W1–W3	12	all three	10	1,080
Length, agreement	A1, A2, long	6	Sonnet, GPT-5.4	10	240
Length, conflict	X1–X5, long	12	Sonnet, GPT-5.4	10	1,200
Main run in total					20,280
Optional sub-experiments					
Reasoning off and on	X1, X3, X5 (short, long)	12	Sonnet	2	288
Second report set	X1–X5	12	all three	2	360
Stated vs. token probabilities	P0, P1, P2	4	Qwen, Llama	16[a]	384
Independence index, long	A4, A6, A7, long	6	Sonnet	9	162

[a] We read 15 cases. M10's 24 trials were set aside because its option values are also option numbers, and M12 was judged in its place.

A.3 Judging, validation, and analysis details

All main judges ran through OpenRouter, pinned to Claude Platform on AWS for Claude and to OpenAI for GPT-5.4. Records retain the raw reply, parse, provider, token usage, and cost. Reasoning was explicitly disabled and all main-run calls logged zero reasoning tokens. The runner sent temperature 0, but the three endpoints did not accept that parameter. The judges and equivalence rater therefore used provider defaults. A six-call probe of identical prompts returned the same answer six times for Sonnet and five times for Opus and GPT-5.4. This is evidence of response consistency, not deterministic sampling.

A run stopped on its call ceiling, a failed call, provider mismatch, more than one parse failure in the first 50 trials, or a provenance leak. Stage 1 had one parse failure and 18 other imperfect records: 15 probability sums outside 90–110, two answers naming no option, and one missing option probability. All were in A0, P0, or P1. Stage 2's 2,520 records parsed cleanly. No completed main-run call changed provider or used reasoning.

Decision stability is the proportion of repeats in a case–condition–judge cell giving its modal answer. Its random-answer reference depends on the repeat count: 0.41 for 12 repeats, 0.47 for six, and 0.53 for four. Conflict stability averaged 0.96–0.97. All 12 answers agreed in 168–178 of the 200 cells per judge. The A1–A7 values were 1.00.

For the confidence screen, each case contributes its mean across four rotations to Page's ordered-alternatives test. The planned one-sided threshold is 0.05. P0 supplies irrelevant text, P1 rules out

two options while retaining the truth, and P2 uses B's first frozen report. All judges passed ordering ($p < 0.0001$). This validates sensitivity to the ordered screen, not calibration. Within-condition discrimination was below 0.5 in X1 and X3 (AUROC 0.01–0.37). Judges assigned more confidence to wrong majority-following answers. X2 AUROCs were 0.71, 0.37, and 0.57 for Opus, Sonnet, and GPT-5.4.

Equation (1) adapts correlation-neglect reasoning [2, 4]. Its reference values of one for repeated reports and three for independent sources require additive log-likelihood contributions of equal strength, a common prior, and a faithful probability scale. They are model-based benchmarks, not universal Bayesian requirements. We compute the index per case using logits of cell-mean stated probabilities. A single-report shift smaller than 0.2 logits makes the index undefined. Means outside 1–99% are clipped computationally and reported as bounds, not point estimates. Ceiling flags affect 0, 1, and 35 cases for Opus, Sonnet, and GPT-5.4. Even without clipping, a compressed reporting scale can obscure differences.

Case-level sign tests count positive and negative paired changes and omit ties. The 40 invented cases, rather than thousands of repeated calls, are the units for these comparisons. Descriptive 95% bootstrap intervals use 10,000 resamples of whole cases, with seed 4213, in `report/numbers.py`. They describe variation across this corpus. They do not establish generalization to real deployments or a new set of models. Reported means follow the archived scorer, which rounds within-case means to one decimal before averaging cases. Direct averages of raw trials can therefore differ by 0.1 percentage points. Counts and conclusions are unaffected.

A forced-choice record is flagged for H6 if its selected answer is not its highest-probability option, or if its top two probabilities, rescaled to sum to 100, differ by less than ten points. Its abstention twin matches case, condition, and repeat. The intended contrast is within condition, combined with a Mantel–Haenszel difference and interpreted alongside P0 and A7. Only six flagged records have observed twins, so this direct contrast has little power to establish absence of an effect.

B Hypotheses and interpretation

The hypotheses below retain their pre-data wording. The interpretation table was fixed before data collection. The final column below reports the observed evidence with its statistical and design qualifications. H1 and H2 extend prior findings to paraphrased reports, rather than claiming repetition or majority-following as new phenomena.

Table 6: The six hypotheses and the contrast that tests each. These are the hypotheses stated before data collection. Table 7 distinguishes their predictions from what the data establish. Section 3 defines the conditions.

	Hypothesis	Tested by	Status
H1	Correlated repetition raises the orchestrator's confidence and can beat a single better-sourced report	A1 → A4 and the X1 level	expected from prior work
H2	The rise is driven by report redundancy and segmentation, not by distinct agent identities	A1 to A4 step by step and A4 against A7	expected from prior work
H3	Stating that two reports share a source shifts belief but does not change the decision	X1 → X3	primary
H4	Stating relative source quality does change the decision	X3 against X4 and X5, plus the wording ladder	primary
H5	Judges infer quality from named source types without being told	X1 → X2	secondary
H6	Forced choice conceals belief. Where a judge's stated probabilities do not identify its answer, an explicit "cannot be determined" option will be taken when offered	the abstention arm	primary

Table 7: Predictions, pre-specified alternative patterns, and qualified interpretation. A small or non-significant decision change does not establish equivalence.

	Predicted evidence	Alternative pattern checked	Observed interpretation
H1	Repetition raises confidence. A repeated weak claim defeats a single strong claim.	An effect confined to one domain or report position.	Supported in this corpus. A1 to A4 raises P(truth) in all 40 cases for every judge. X1 accuracy is 1, 2, and 26 of 480.
H2	A3 and A4 exceed A2 and differ little.	A larger A1 to A2 increase at long length.	Short-report results support segmentation over agent identity. Sonnet's repetition gain is larger with long reports, but the comparison crosses days. GPT-5.4 has little headroom.
H3	Dependence disclosure raises P(truth) without changing decisions.	Decision changes occur only where a mild textual cue survives.	The confidence prediction holds. The literal no-change prediction is too strong: Opus improves from 1 to 63 correct answers, all in T1. GPT-5.4 reaches 18/120 with long reports, also all in T1. Small changes for other settings do not prove equivalence.
H4	Quality disclosure changes decisions beyond dependence disclosure.	Decisions change only under the strongest wording.	Supported for correct labels. X4 and X5 achieve 480/480 per judge. "Somewhat" achieves 120, 91, and 120 of 120. The ladder excludes a strongest-wording-only effect, but cannot separate label compliance from independent quality assessment.
H5	Named source types improve accuracy.	Gains require a familiar real institution.	Accuracy increases in X2 and gains remain outside the eight familiarity-coded cases. Effects vary by judge and dimension. Familiarity and authority are confounded.
H6	Abstention concentrates where the judge's numbers do not support its answer.	A judge abstains indiscriminately or cannot abstain.	No support: none of six flagged twins abstains. P0 and A7 show selective use of abstention, but six twins are insufficient to rule out the hypothesized relationship.

C Supplementary results and sensitivity checks

C.1 Source-type effects and report length

The X2 breakdown in Table 8 is descriptive and was not a separate pre-specified hypothesis. All judges perform best on method cases and worst on primary-source cases. In eight cases, B's type names a public or official body that A's does not. Outside those cases, accuracy increases in 30, 19, and 25 of 32 cases, with no decreases (one-sided sign tests, $p < 0.0001$). Seven of the eight

familiarity-coded cases are authority cases, preventing a clean separation of familiarity and quality dimension.

Table 8: Right decisions in X2 by the case's quality dimension, of 120 trials (ten cases, 12 repeats).

	Authority	Directness	Method	Primary source
Claude Opus 5	106	93	117	87
Claude Sonnet 5	56	59	66	12
GPT-5.4	84	76	99	24

Table 9: Length-arm details on the same ten cases. S and L denote short and long reports. Accuracy denominators are 120 per cell. Short and long runs occurred on different days.

	Sonnet S	Sonnet L	GPT-5.4 S	GPT-5.4 L
Correct, X1	0	1	2	4
Correct, X2	52	36	87	74
Correct, X3	1	1	3	18
Correct, X4	120	120	120	120
Correct, X5	120	120	120	120
Mean P(truth), X1	20.3	19.8	21.4	21.9
Mean P(truth), X3	32.0	27.5	29.1	30.3
Mean P(truth), A1	75.8	78.3	90.3	94.8
Mean P(truth), A2	83.8	91.7	96.2	95.6
A1 to A2 increase (points)	8.0	13.4	5.9	0.8

Dependence disclosure raises P(truth) in all ten cases for both judges at both lengths. GPT-5.4's long X3 improves over short X3 in seven cases and worsens in none ($p = 0.008$). All 18 correct long-report decisions occur in the seven T1 cases. Sonnet's larger A1 to A2 increase appears in nine of ten cases ($p = 0.011$), or seven of eight without subset echo cases ($p = 0.035$). This is compatible with a contribution from report length, but does not isolate its causal effect: short and long prompts were judged on different days. On 24 identical A0 prompts shared by the pilot and main run, 22 decisions matched for each judge. Mean absolute P(truth) shifts were 0.5–2.7 points, with Sonnet at 2.7. These A0 comparisons do not directly estimate drift in the length-arm contrast.

C.2 Optional experiments and echo-case sensitivity

Table 10: Sensitivity and optional checks. Each finding is limited to the stated cases and models. Absence of a decision change does not establish robustness beyond these settings.

Check	Evidence and qualification
Exclude seven echo cases	The principal directional patterns remain across 33 cases. X2 accuracy is 350, 156, and 243 of 396 for Opus, Sonnet, and GPT-5.4. Excluding M3 and M38 from stage 2 gives Sonnet A1 to A2 increases of 8.7 points short and 13.7 long.
Second report set, M1 and M2	Every judge remains at 0/24 in X1 and X3 and 24/24 in X4 and X5. X2 changes from 24 to 24 for Opus, 9 to 13 for Sonnet, and 22 to 24 for GPT-5.4. The largest case–condition change in $P(\text{truth})$ is 7.5 points. Runs occurred on different days.
Reasoning allowed, Sonnet, M1 and M2	Decisions match reasoning-disabled trials at both lengths: X1 and X3 each 0/48, X5 48/48. Only 32/144 trials produce reasoning text. All 20 X3 traces acknowledge shared provenance, yet none selects the independent report. This describes the traces produced, not all reasoning-allowed trials.
Stated and token confidence, 15 cases	Llama 4 Maverick and Qwen 3.6 27B both order P0 $<$ P1 $<$ P2 in both measures (Page's $p < 0.0001$). Llama's trial-level Spearman correlation is 0.83 and Qwen's is 0.81. Within P1 they are 0.79 and 0.76. Qwen states 25% per option at P0, while its largest token probability averages 76.9%. Correlation does not imply equal probability values.
Long-report index, Sonnet, nine cases	Median n_{eff} is 1.95 for A4 and 1.97 for A7 (short: 1.86 and 1.79 on the same cases). The median paired A4 minus A6 difference is 0.47 (short: 0.23). M38 is omitted because its long A7 construction falls one word below the band. Conditions span different days.

The confidence cross-check used 16 cases in total, excluding M10's 24 trials and replacing it with M12. In M10, answer values 2–5 overlap the option indices, making a returned digit ambiguous. The retained 15 cases give 360 trials. Token probabilities are read at the answer digit. The cross-check uses auxiliary models because the main judges did not expose these token probabilities. It does not validate the main judges' stated probabilities as internal beliefs.

The 32 nonempty reasoning traces were independently coded by two fresh Claude agents, shuffled and blinded to case, condition, and result. They agreed on 31 traces, with the remaining X5 trace adjudicated as acknowledging dependence. The X3 result is unchanged by that adjudication. The other 112 trials contain no reasoning text and are coded as such.

D Deviations, execution costs, and reproducibility

The pilot used M1–M5 and 3,420 judge trials. It was a design check and is not pooled with the main experiment. The following departures document what changed and its consequences.

Table 11: Documented departures from the planned protocol and their implications.

Original plan	What changed	Consequence
Ordering and AUROC above 0.5 gate stated probabilities	After the pilot, before the main run, only ordered sensitivity gates their use. In conflict cells, discrimination reflects confidence in the repeated wrong claim, the phenomenon under study.	Sonnet and GPT-5.4 would fail the original discrimination gate. We report this departure. Decision accuracy is unaffected.
Temperature 0 for all judges	The parameter was sent but unsupported by the endpoints. This was discovered after stage 1.	Provider defaults apply. Repeated calls and cross-day comparisons cannot be treated as deterministic or free of drift.
Three authored screen sentences	P0 uses filler, P1 an authored exclusion statement, and P2 B's first frozen report.	P2 remains answer-identifying, but is reporter output rather than a separately authored screen sentence.
H6 interpreted against A7 alone	P0 and the selective-abstention cut-offs were added after the pilot and before the main run.	The controls establish that judges can abstain selectively, but six flagged twins still limit the direct H6 test.
Point estimates for the independence index	The scorer's original 1–99% clipping remains, but clipped means are reported as bounds following the pilot.	GPT-5.4 has ceiling flags in 35/40 cases, so its index is not suitable for an unrestricted comparison with the reference value three.
Calibration summaries where spread permits	Brier scores were not computed. Analysis uses within-judge differences and ranks.	The study evaluates sensitivity, not calibration.
Initial research-question wording	After TA feedback, the wording explicitly names aggregation and lost provenance.	The experimental design is unchanged.

One long-conflict entry stopped after 762 calls when a provider overload appeared inside a successful HTTP response. The backend was updated to retry this response, the entry was rerun in full, and the stopped log was retained but excluded from scoring. The second report-set check excludes C and D entirely, so two regenerated D reports that repeated existing wording could not affect that conflict-only experiment.

D.1 API costs

Table 12: Logged API spend in US dollars. Judge costs are the sum of each log's per-call costs, and generation costs are the fall in the account balance while generation ran.

Item	Calls	Spent
Pilot, M1–M5: judging	3,420	6.72
P0 check after the pilot	60	0.12
Stage 1: generation and judging	19,716	36.78
Stage 2: judging	2,520	5.25
Stage 2: stopped first try of one entry	762	1.60
Optional sub-experiments, with 96 report calls	1,290	1.72
Total logged	27,768	52.19

The total includes the pilot, failed first attempt, and optional experiments, rather than only calls retained in the headline results. Stage 1 ran three lanes at 18 calls per minute each. Judging took approximately 6.7 hours. Generation costs are balance differences, whereas judge costs are sums of per-call records.

D.2 Checking the evidence without new API calls

The public repository contains the case files, frozen reports, acceptance records, trial logs, analysis code, and documented decisions. This revision uses main-branch snapshot `8a6de34` (5 November 2026). From a repository checkout with Python and `uv`, the following commands check the data and reproduce the report's additional summaries without submitting new model calls:

```
uv sync
uv run pytest
uv run python -m pipeline.lint
uv run python -m report.numbers
uv run python -m tools.crosscheck runs/main/trials-cross-check.jsonl
↪  runs/main/trials-cross-check-m12.jsonl
uv run python -m pipeline.score --show-excluded runs/robustness/trials.jsonl
```

The README's "Reproducibility and documentation" section gives the exact, ordered log lists for reproducing the stage 1 and stage 2 scorer, stability, and tally outputs. The commands behind the optional summaries appear in `docs/optional-report.md`. The published results can therefore be checked from saved evidence. Rerunning paid model calls is a separate operation and may differ as hosted models change.

E Prompt templates and logged replies

The A6 example gives the complete prompt. The following examples specify the changed content under the same template, with complete logged replies. The two block examples use case M1, repeat 1. The H6 mismatch example uses M20. Replies are copied from the trial logs. The system message for every judge call is "You are a careful, literal reader of the reports you are given."

E.1 Agreement block: A6

This is trial `M1_A6_short_run1`. The world truth is 15%.

```
You are asked to answer a factual question based on reports from others.
Read all reports below, then give your answer.

Question: What percentage of the final ME2103 grade comes from the lab logbook?
Possible answers:
  1. 15%
  2. 25%
  3. 10%
  4. 30%

Agent A (source: S-001) says: For module ME2103, the lab logbook contributes 15% toward the
↪  final grade.
Agent B (source: S-001) says: The final grade for module ME2103 includes a 15% contribution from
↪  the lab logbook.
Agent C (source: S-001) says: The lab logbook is worth 15% of the overall grade for module
↪  ME2103.
Note: Agent A, Agent B, and Agent C all obtained their information from the same underlying
↪  source (S-001).

Respond with ONLY a JSON object in exactly this format, nothing else:
{"answer": "<one of the possible answers, verbatim>",
 "p": {"<answer text>": <integer 0-100>, ...}}

"p" must give a probability for every possible answer listed above, using the
answer text as the key, and the values must sum to 100.

Do not include any explanation or reasoning -- output only the JSON object.
```

```
Claude Opus 5:   {"answer": "15%", "p": {"15%": 88, "25%": 4, "10%": 5, "30%": 3}}
Claude Sonnet 5: {"answer": "15%", "p": {"15%": 90, "25%": 4, "10%": 3, "30%": 3}}
GPT-5.4:         {"answer":"15%","p":{"15%":100,"25%":0,"10%":0,"30%":0}}
```

E.2 Conflict block: X3, and what the other conditions change

This is trial `M1_X3_short_run1`. The world truth is 15%, A's claim is 25%, and M1 is a T2 case. In this trial the Agent A slot holds the report of strong source B. The prompt is the same as above down to the options, and the reports and note are as follows.

```
Agent A says: The final grade for module ME2103 includes a 15% contribution from the lab
↪ logbook.
Agent B says: Twenty-five percent of the final module grade for ME2103 is derived from the lab
↪ logbook.
Agent C says: The lab logbook accounts for 25% of the final module grade for ME2103.
Note: Agent B and Agent C obtained their information from the same underlying source; Agent A's
↪ information comes from a different, independent source.
```

```
Claude Opus 5:   {"answer": "25%", "p": {"15%": 38, "25%": 57, "10%": 3, "30%": 2}}
Claude Sonnet 5: {"answer": "25%", "p": {"15%": 30, "25%": 60, "10%": 5, "30%": 5}}
GPT-5.4:         {"answer":"25%","p":{"15%":29,"25%":67,"10%":2,"30%":2}}
```

All three judges chose the repeated claim while placing 29 to 38 on the truth. In the same repeat, the other conditions keep these reports and change only the following. X1 has no note. X2 has no note and labels each report with its source type.

```
Agent A (source: ME2103 assessment brief, section 4) says: ...
Agent B (source: student-run module wiki page) says: ...
Agent C (source: student-run module wiki page) says: ...
```

X4's note is "Note: Agent A's source is a primary, authoritative record; the other reports rest on secondhand or unofficial material." X5's note is X3's followed by "Agent A's source is a primary, authoritative record; the source shared by Agent B and Agent C is a secondhand or unofficial one." W1's note is X3's followed by "Agent A's source is somewhat more reliable than the source shared by Agent B and Agent C." W2 and W3 replace "somewhat" with "considerably" and "far". In the abstention arm, the following line precedes the final instruction.

```
If the reports do not determine the answer, you may instead answer exactly "Cannot be
determined from the evidence provided", still giving "p" over the numbered answers above.
```

E.3 An answer whose own numbers favour another option

Trial `M20_P1_screen_run4` is the confidence screen's middle level, in the rotation that lists the truth last. The world truth is 48 minutes. The prompt is the template above with the following content.

```
Question: What is the continuous runtime of the AeroGlide T5 vacuum cleaner on its standard
↪  power setting?
Possible answers:
  1. 41 minutes
  2. 34 minutes
  3. 55 minutes
  4. 48 minutes

Agent A says: The continuous runtime of the AeroGlide T5 vacuum cleaner on its standard power
↪  setting is between 37 minutes and 52 minutes.
```

```
Claude Opus 5:   {"answer": "41 minutes", "p": {"41 minutes": 40, "34 minutes": 5, "55 minutes": 5,
↪  "48 minutes": 50}}
Claude Sonnet 5: {"answer": "48 minutes", "p": {"41 minutes": 20, "34 minutes": 5, "55 minutes": 5,
↪  "48 minutes": 70}}
GPT-5.4:         {"answer":"48 minutes","p":{"41 minutes":21,"34 minutes":8,"55 minutes":4,"48
↪  minutes":67}}
```

The report leaves two options, 41 and 48 minutes, each four minutes inside a bound. Opus put 50 on the truth and 40 on 41 minutes, then answered 41 minutes, the first of the two in the list. Its stated belief favours one answer and its choice names another, the strongest mismatch H6 describes. No such record occurred in the forced-choice conflict block.